

Estimating spatial firm productivity spillovers using UK microdata

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Declaration

I, Aloysius Lip, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

I declare the use of GenAI tools in an assistative role (UCL Category 2) for this submission. I used Gemini 3.10 Pro (Google DeepMind, February 2026. <<https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-1-Pro-Model-Card.pdf>>). This is limited to the roles of: supporting the particular process of reviewing and refactoring code (data analysis), as a support tutor (model and empirical approach).

Name: Aloysius Lip

Date: 2nd September, 2026

Classification

This piece of research is primarily (please indicate as appropriate):

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

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Table of Contents

Literature	1
Data	1
Model	3
Results	3
Results	5
Bibliography	8
Appendix 1	9
Appendix 2	10
Appendix 3	11

List of Figures

Abstract

~300 word summary of your research topic, methodology and findings

Acknowledgements

(if applicable)

Word count:

Literature

My investigation is rooted in Marshall (1920)'s taxonomy, which identifies three channels to productivity spillovers: (1) input-output costs, (2) labour pooling, and (3) knowledge spillovers. There is no reason to think these are exhaustive. We might theorise that there may be competitive effects and cultural factors. Krugman also labels all these channels as 'centripetal' to an urban core; there are likewise 'centrifugal' factors that push firms away from the urban core, such as destructive (zero-sum) competition, congestion, and saturation costs. A further extension could develop models on these further channels and empirically test them. Furthermore, these productivity spillovers are incredibly salient to thinking about production structures.

In this investigation, I concentrate on (3) knowledge spillovers, tackling a difficult identification problem highlighted in Glaeser (2001). This is more relevant to the knowledge-based services economy that dominates the UK. Input-output costs and labour-pooling matter more for industrial manufacturing firms, especially the latter which was devised in response to the phenomenon emergence of industry towns.

Most agglomeration literature bundles all three channels together. Glaeser (2010) and Krugman (2010) identify how vital identifying knowledge spillovers are. That need to estimate them well only grows with the modern, knowledge economy of the UK. (Goldin, 2023) speculates that agglomeration effects have "increasing strength" attributable to the way in which "knowledge workers in particular become more productive when located near to one another. Information may be more available than ever, but the evidence suggests that it still tends to flow largely through in-person networks".

Specific approaches to investigating knowledge spillovers via innovation, instrumenting via patent publications comes from US evidence, including (Jaffe, 1993), (Berkes and Ruben, 2021) and more recently (Bloom, Schankerman and Van Reenen, 2013).

Knowledge spillovers have been a well-evidence factor of within-firm managerial-derived productivity gains. A field-starting MIT study from Allen (1977) observed an exponentially declining curve depicting the frequency of communication between two individuals against their physical distance in an office.

Data

I have used data from FAME for this investigation. FAME is a comprehensive set of financial datapoints from UK firms, with instruments appropriate for this analysis, in particular number of employees and balance sheet data from which economically interesting variables can be derived. FAME is provided by Moody's, an international credit agency, and collects its data from UK firms (via survey).

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from FAME in February 2025 by Professor Lars Nesheim at UCL for (Irastorza-Fadrique et al., 2026). This project stored a series of Excel tables splitting financial balance sheet variables by their category. Downloadable indicators under 'a1_ID' and 'a5_misc' contain fixed datapoints about each firm; 'a2_key_finance', 'a3_assets', and 'a4_profits' contain yearly financial data about each firm. I designed a series of Python scripts to ingest this data and store it in a 'DuckDB' database, separated into fixed and yearly datapoint tables. 'DuckDB' is an open-source, local, relational (SQL) architecture, designed for processing bulk queries across

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columns of variables, making it well-suited for econometric data analysis.

Model

I start by estimating a basic 2-factor production model, in line with Solow's productivity decomposition. Capital is calculated from FAME's `fixed_total` variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is . Labour is taken from the number of firm employees from the FAME balance sheet.

Insert
capital-
output
ratio

Results

The result of my LLM model are as follows.

Parameter	peer3	peer3_no_firm_fe	peer3_employees
β_0	2.075	0.310	2.035
const	(0.044)***	(0.038)***	(0.047)***
β_1	2.075	0.310	2.035
const	(0.044)***	(0.038)***	(0.047)***
β_2	0.033	0.204	0.033
peer_tfp_ttw_a_donut	(0.013)***	(0.013)***	(0.013)***
β_3			0.009
ln_employees			(0.004)**
β_4	0.015	0.142	0.015
peer_tfp_pc4_donut	(0.005)***	(0.008)***	(0.005)***
β_5	0.280	0.559	0.280
peer_tfp_pc8	(0.005)***	(0.004)***	(0.005)***
Observations	764,537	764,537	764,537
i	107,143		107,143
t	19	19	19
$R^2(within)$	0.040	-0.000	0.040

The results of my distance decay model are as follows.

Parameter	dd_gravity1	dd_neg2	dd_expdd3	dd_struct1	dd_struct2	dd_struct3
β_0	2.479	2.586	2.417	2.698	2.799	2.652
const	(0.029)***	(0.029)***	(0.032)***	(0.033)***	(0.032)***	(0.035)***
β_1	0.168			0.168		
tfp_wav1	(0.003)***			(0.003)***		
β_2		0.135			0.135	
tfp_wav2		(0.003)***			(0.003)***	
β_3			0.184			0.183
tfp_wav3			(0.005)***			(0.005)***
β_4	-0.009	-0.009	-0.009	0.619	0.619	0.619
ln_employees	(0.004)**	(0.004)**	(0.004)**	(0.005)***	(0.005)***	(0.005)***
β_5	0.013	0.012	0.016			
ln_nb_peers	(0.004)***	(0.004)***	(0.004)***			
β_6				0.265	0.265	0.265
ln_total_assets				(0.004)***	(0.004)***	(0.004)***
Observations	1,067,013	1,067,013	1,067,013	1,067,013	1,067,013	1,067,013
i	139,162	139,162	139,162	139,162	139,162	139,162
t	19	19	19	19	19	19
$R^2(within)$	0.010	0.010	0.004	0.392	0.391	0.388

The results of my dynamic panel are as follows

Parameter	dyn1	dyn2	dyn3
β_0	1.371	1.577	1.064
const	(0.134)***	(0.144)***	(0.103)***
β_1	0.005	0.005	0.005
L1.ln_gva1	(0.001)***	(0.001)***	(0.001)***
β_2	0.696	0.689	0.709
ln_employees	(0.038)***	(0.037)***	(0.040)***
β_3	0.259	0.266	0.247
ln_total_assets	(0.041)***	(0.041)***	(0.044)***
β_4	0.503		
tfp_wav1	(0.028)***		
β_5		0.423	
tfp_wav2		(0.023)***	
β_6			0.618
tfp_wav3			(0.042)***
Hansen Test	X (0.417)	X (0.459)	X (0.246)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.746)	X (0.600)	X (0.945)
Observations	713,339	713,339	713,339
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tfp_wav3			(0.005)***			(0.005)***
β_4	-0.009	-0.009	-0.009	0.619	0.619	0.619
ln_employees	(0.004)**	(0.004)**	(0.004)**	(0.005)***	(0.005)***	(0.005)***
β_5	0.013	0.012	0.016			
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i	139,162	139,162	139,162	139,162	139,162	139,162
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Appendix 1

Appendix 2

Appendix 3